

Rings and Modules

Prelude

1.

Find an example of each of the following.

- (a) * A short exact sequence of modules that is not split exact.
- (b) A projective module that is not free.
- (c) A flat module that is not projective.
- (d) * A flat $\mathbb{Z}$ -module that is not free.
- (e) Nonzero modules M, N over a ring R such that $M \otimes_R N = 0$.
- (f) A ring R and an R -algebra A that is finitely generated as an algebra over R but is not finitely generated as an R -module.

* 2.

Let $M = \mathbb{C}[x]/(x^2 + x)$ and $N = \mathbb{C}[x]/(x - 1)$. Compute the dimension of the following tensor products as vector spaces over $\mathbb{C}$:

- (a) $M \otimes_{\mathbb{C}[x]} N$
- (b) $M \otimes_{\mathbb{C}} N$
- (c) $M \otimes_{\mathbb{C}[x^2]} N$.

Fugue

* 3.

Let R be a commutative ring.

- (a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.
- (b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

4.

Show that if R is a PID (principal ideal domain) and M is an R -module, then M is flat if and only if M is torsion free.

* 5.

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

* 6.

Let R be a commutative ring.

- (a) Give an example of a free R -module M and an injective homomorphism $M \rightarrow M$ which is not an isomorphism. (For this part, R can be whatever commutative ring you like.)
- (b) If M_1 and M_2 are free of finite ranks r_1 and r_2 , show that a surjective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \geq r_2$.
- (c) Assume furthermore that R is an integral domain. Show that an injective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \leq r_2$.

* 7.

Let I and J be ideals in a commutative ring R .

- (a) Show that if $I + J = R$, then $I \cap J = IJ$.
- (b) Suppose that I and J are ideals in $\mathbb{C}[x]$, and suppose that $I + J = (x)$. Show that $(I \cap J)/IJ$ is 1-dimensional as a complex vector space, and moreover that it is isomorphic to $\mathbb{C}[x]/(x)$ as a $\mathbb{C}[x]$ -module.
- (c) On the other hand, for general commutative rings R , once $R/(I + J)$ is not trivial, the difference between $I \cap J$ and IJ can be large even if $R/(I \cap J)$ is small. Demonstrate this by showing that, if $R = \mathbb{C}[x, y]$, then there exists ideals I and J in R such that $I + J = (x, y)$ and the dimension of $(I \cap J)/IJ$ as a $\mathbb{C}$ -vector space is at least 100.

* 8.

(This problem has been slightly modified from its original form.) Let $B = \mathbb{Q}[x, y]/(xy)$.

- (a) Consider the map $\mathbb{Q}[t] \rightarrow B$ sending t to x . This map makes B into a $\mathbb{Q}[t]$ -module. Show that B is *not* a finite $\mathbb{Q}[t]$ -module under this map.
- (b) Consider the map $\mathbb{Q}[t] \rightarrow B$ sending t to $x + y$. Show that this map makes B into a finite $\mathbb{Q}[t]$ -module.
- (c) Consider B as a $\mathbb{Q}[t]$ -module under the map from part (a). Is B a flat $\mathbb{Q}[t]$ -module? What about when considered as a $\mathbb{Q}[t]$ -module under the map from part (b).